# Safety & ergonomics

National 5 · Design · Worksheet

D&M WITH MR KELLY

Name _____ Class _____ Date _____

**By the end of this worksheet I can…**

- ☐ describe how ergonomics makes products safe, comfortable and easy to use
- ☐ use anthropometric data and choose the right percentile
- ☐ describe how safety has influenced the design of a product

## I DO — Worked example

Read how a full-mark answer is built

*A cordless hand vacuum cleaner.*

**1 Describe how ergonomics has influenced the design of the hand vacuum cleaner.** (4 marks)

**Full-mark answer** each ✓ is one mark

The handle diameter is sized to fit the grip of an adult hand ✓. The power button is placed under the thumb so it can be pressed while holding the handle ✓. The vacuum is light so it can be held up to clean shelves without the arm tiring ✓. The bin release catch is coloured red so users can see where to press ✓.

**Why answers lose marks**

*"It uses the 95th percentile so it is comfortable and easy to use."*

0 marks: percentiles on their own and generic "comfortable/easy to use" are not credited.

## WE DO — Guided practice

Use the starters and word bank to help you

*A child's scooter.*

**2 Describe how safety has influenced the design of the scooter.** (3 marks)

**Sentence starters**

First, …
Next, … using a …
Then …

**Word bank**

non-slip · stability · operated · rounded · wheelbase · folding · rubber · handlebars

Tip: 3 marks = 3 separate, developed points.

**YOU DO** **On your own** No notes. Check your answers afterwards.

### 3 Quick check: tick one box for each.

**a)** Which feature mainly improves safety?
☐ Brand logo ☐ Guard over moving blades ☐ Glossy finish ☐ Secondary storage

**b)** A bright red, clearly labelled off switch reassures the user. This is...
☐ Psychology ☐ Anthropometrics ☐ Physiology ☐ Function only

**c)** "The force needed to squeeze the trigger is low so the user does not tire" is an example of...
☐ Anthropometrics ☐ Physiology ☐ Psychology ☐ Aesthetics

**d)** Which percentile is used to position an emergency stop button?
☐ 5th ☐ 50th ☐ 95th ☐ 99th

### 4 Key terms: write the letter of the matching description next to each term.

| | | |
|---|---|---|
| 1. **Anthropometrics** | ____ | **A** How a product's look, feel, sound, smell or taste affects the user. |
| 2. **Physiology** | ____ | **B** Strength, effort, movement and strain required to use a product. |
| 3. **Psychology (ergonomics)** | ____ | **C** The study of human body measurements used to size products. |

### 5 Explain which percentile is used for reach, and why. (2 marks)

### 6 Explain which percentile is used for clearance, and why. (2 marks)

**Check and reflect**
Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?